

Scribe 2 (Part 2): Questions 4–6

Implementation within the Simulation Pipeline

- Describe how the randomized algorithm has been integrated into your existing simulation pipeline.
- What changes were made to the code or system architecture?
- How is randomness generated or sampled in the implementation?
- Does the rest of the pipeline remain the same as the baseline?

Simulation Pipeline Implementation

Integration of the Randomized Algorithm

The randomized algorithm was incorporated into the existing simulation pipeline by transforming the deterministic routing system into a stochastic one through probabilistic modeling of travel times.

In the baseline implementation, route selection was performed using the A* algorithm, where edge weights were fixed deterministic travel costs. In the enhanced approach, each road segment is modeled as a random variable, allowing the system to capture uncertainty in traffic conditions.

Instead of evaluating a single deterministic route, the improved method estimates the distribution of total route travel time by repeatedly executing stochastic simulations (Monte Carlo sampling). The decision-making process is then based on the probability of arriving on time, rather than simply selecting the route with minimum expected travel time.

Code and System Architecture Changes

The existing system was modified in the following ways:

- **Edge Cost Representation (ECR):** Deterministic edge weights based on distance were replaced with stochastic travel times modeled using lognormal distributions.
- **Route Evaluation Mechanism:** The single-pass A* evaluation was extended to a multi-run simulation framework, where travel times are sampled repeatedly and aggregated to estimate route performance.
- **Decision Rule Update:** The routing objective was changed from minimizing total travel time to maximizing the probability of meeting a predefined deadline.
- **Metric Computation:** Additional statistical measures such as variance, percentile travel time, and reliability were introduced to evaluate route performance under uncertainty.

Randomness Modeling

Randomness in the system involves three key aspects: randomness generation, randomness sampling, and randomness testing (Kimura et al., 1994).

Probabilistic sampling is used to model segment travel times. Each segment travel time is assumed to follow a lognormal distribution:

$$T_i \sim \text{Lognormal}(\mu_i, \sigma_i)$$

For each simulation run, independent samples are drawn for all segments along a route, and the total travel time is computed as:

$$T_{\text{route}} = \sum_{i=1}^n T_i$$

This process is repeated multiple times to approximate the empirical distribution of route travel time. The probability of on-time arrival is then estimated as the fraction of simulations in which:

$$T_{\text{route}} \leq D$$

Alignment with Baseline Pipeline

The project remains consistent with the baseline pipeline, as it is designed to achieve the same routing objectives. Core components such as road network representation, route generation algorithms, and the simulation environment remain unchanged.

However, the key modification lies in the evaluation and decision-making process. Deterministic calculations are replaced with probabilistic analysis, enabling more robust and uncertainty-aware routing decisions. Thus, the randomized algorithm augments the existing pipeline rather than replacing it.

Summary

Randomness transforms the routing framework from a deterministic shortest-path model into a probabilistic decision-making system. By incorporating stochastic travel time modeling and Monte Carlo simulation, the system achieves greater reliability and provides a more realistic representation of traffic dynamics.

Evaluation Metrics and Comparative Results

- Identify the evaluation metrics used to measure system performance.
- Why are these metrics appropriate for your problem domain?
- How does the randomized algorithm perform compared to the deterministic baseline?
- Present and interpret the results of this comparison.

Evaluation Metrics and Performance Analysis

1. Evaluation Metrics

System performance was evaluated using the following metrics:

- **Probability of On-Time Arrival:**

$$P(T_{\text{route}} \leq D)$$

Determines the probability that the destination is reached within the deadline.

- **Expected Travel Time:**

$$\mathbb{E}[T_{\text{route}}]$$

Represents the mean travel time across simulations.

- **Variance of Travel Time:**

$$\text{Var}(T_{\text{route}})$$

Measures variability and uncertainty in travel time.

- **Percentile-Based Metrics (e.g., 95th percentile):** Indicates worst-case travel time under high-confidence conditions.
- **Confidence Intervals:** Provide probabilistic guarantees on estimated travel times.

2. Appropriateness of Metrics

- **Reliability-Focused Objective:** Since the problem involves uncertainty in traffic conditions, probability of on-time arrival is more meaningful than average travel time.
- **Capturing Risk and Variability:** Variance and percentile metrics capture delay risks that deterministic approaches cannot model.
- **Decision-Oriented Evaluation:** These metrics align with real-world scenarios where consistency and reliability are more important than ideal average performance.
- **Alignment with Concept Map:** The metrics directly relate to probabilistic modeling components such as PDF, CDF, and deadline comparison.

3. Randomized vs Deterministic Comparative Performance

- **Deterministic Baseline (A*):**
 - Performs shortest-path optimization using fixed edge weights.
 - Does not account for variability and uncertainty.
 - Often selects routes with low mean but high variance.
- **Randomized Strategy (Monte Carlo):**
 - Models travel time as a stochastic variable.
 - Evaluates routes using repeated simulations.
 - Maximizes the probability of meeting deadlines.
 - Avoids high-variance, high-risk routes.

4. Results and Interpretation

- **Enhanced Reliability:** Experimental results show that the deterministic A* path has moderate on-time probability (e.g., 66.8%), whereas the Monte Carlo approach achieves significantly higher reliability (e.g., 99.4%).
- **Effect of Variance:** The randomized method prefers routes with lower variance, even if their expected travel time is slightly higher, resulting in more predictable outcomes.
- **Risk-Aware Routing:** The results demonstrate that shortest paths are not always optimal under uncertainty due to heavy-tailed delay distributions.
- **Shift in Optimization Perspective:**

$$\min \mathbb{E}[T] \rightarrow \max P(T \leq D)$$

- **Trade-Off:** While Monte Carlo improves reliability, it introduces additional computational overhead compared to deterministic methods.

5. Summary

- The evaluation metrics effectively capture both performance and uncertainty.
- The randomized algorithm significantly outperforms the deterministic baseline in uncertain and congested environments.
- The results highlight the importance of incorporating variance and probabilistic reasoning in routing decisions.

Insights and Remaining Challenges

Based on your experiments:

- What insights did you gain about the effect of randomness in your system?
- In what situations does the randomized strategy perform better or worse than the baseline?

1. Randomness Effect in the System

- **Modeling Uncertainty:** Randomness enables the system to model travel time as a stochastic variable rather than a deterministic value, reflecting real-world traffic variability.
- **Capturing Variability and Extreme Events:** The probabilistic model accounts for skewness and rare but significant delays that deterministic models fail to capture.
- **Risk-Aware Decision Making:** The system evaluates routes not only based on expected travel time but also on variance and distribution characteristics.
- **Shift in Optimization Objective:** The objective shifts from minimizing expected travel time to maximizing reliability:

$$P(T_{\text{route}} \leq D)$$

- **Distribution-Based Evaluation:** Randomness allows estimation of full probability distributions (PDF/CDF), enabling calculation of confidence intervals and percentile-based metrics.

2. Cases Where the Randomized Strategy Performs Better

- **High Variability Environments:** Monte Carlo methods outperform deterministic routing in congested or unpredictable traffic conditions.
- **Deadline-Constrained Scenarios:** The randomized strategy increases the probability of meeting strict deadlines.
- **Variance-Sensitive Routing:** Routes with slightly higher mean but lower variance are preferred for better reliability.
- **Enhanced Reliability Measures:** This approach provides higher probabilities of on-time arrival compared to deterministic methods.

3. Cases Where Deterministic Baseline Performs Better

- **Low Variability Conditions:** In stable or uncongested environments, both approaches perform similarly, making randomness unnecessary.
- **Computational Constraints:** Deterministic algorithms like A* are more computationally efficient, whereas Monte Carlo methods introduce overhead.
- **Objective Distance-Based Optimization:** When minimizing distance or fuel consumption is the sole objective, deterministic routing is more suitable.
- **Real-Time Constraints:** Randomized approaches may not be feasible in time-sensitive systems due to the need for repeated simulations.

4. Remaining Challenges

- **Integration into Search Process:** Currently, stochastic evaluation is applied after route generation. Incorporating randomness directly into graph traversal remains a challenge.
- **Scalability:** Large-scale networks require significant computational resources to perform multiple simulations.
- **Dynamic Environment Adaptation:** The model assumes fixed probability distributions; real-time updates and adaptive re-routing are not yet implemented.
- **Model Accuracy:** The accuracy of results depends heavily on how well the probability distributions represent real travel times.

5. Summary

- Routing is transformed from a deterministic optimization problem into a probabilistic decision-making framework.
- Reliability-based routing considers both mean and variance.
- Monte Carlo methods are especially effective in uncertain and time-sensitive environments.
- Computational cost and lack of dynamic integration remain key limitations.

End of Submission